Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Piecewise Functions	Made up of 2 or more functions, each with its own domain
Evaluating Piecewise Functions	**Directions:** Find each value given the piecewise function.

1. $f(x) = \begin{cases} 2|x-5| & \text{if } x < -3 \\ -x^2 + 4x & \text{if } -3 \le x < 7 \\ \sqrt{2x+6} & \text{if } x \ge 7 \end{cases}$

a) $f(-5)$

$2|-5-5|$

$2|-10| = \boxed{20}$

b) $f(2)$

$-(2)^2 + 4(2)$

$-4 + 8 = \boxed{4}$

c) $f(7)$

$\sqrt{2(7)+6}$

$\sqrt{20} = \boxed{2\sqrt{5}}$

2. $g(x) = \begin{cases} \frac{1}{2}x^3 + 7 & \text{if } x \le -4 \\ 8 & \text{if } -4 < x < -2 \\ [\![x-4]\!] & \text{if } x \ge -2 \end{cases}$

a) $g(-3)$

$\boxed{8}$

b) $g(-8)$

$\frac{1}{2}(-8)^3 + 7$

$\frac{1}{2}(-512) + 7$

$= \boxed{-249}$

c) $g(3.1)$

$[\![3.1 - 4]\!]$

$[\![-.9]\!]$

$= \boxed{-1}$

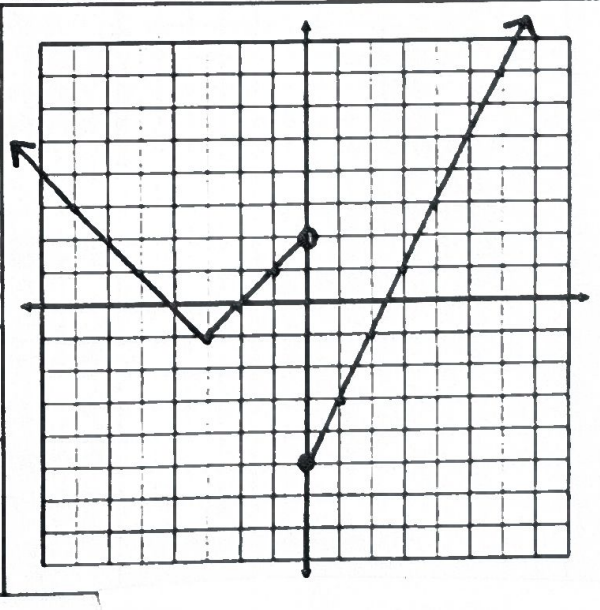
Graphing Piecewise Functions	**Directions:** Graph each function, then give the domain, range, and identify the location and type of any discontinuities.

3. $f(x) = \begin{cases} |x+3| - 1 & \text{if } x < 0 \\ 2x - 5 & \text{if } x \ge 0 \end{cases}$

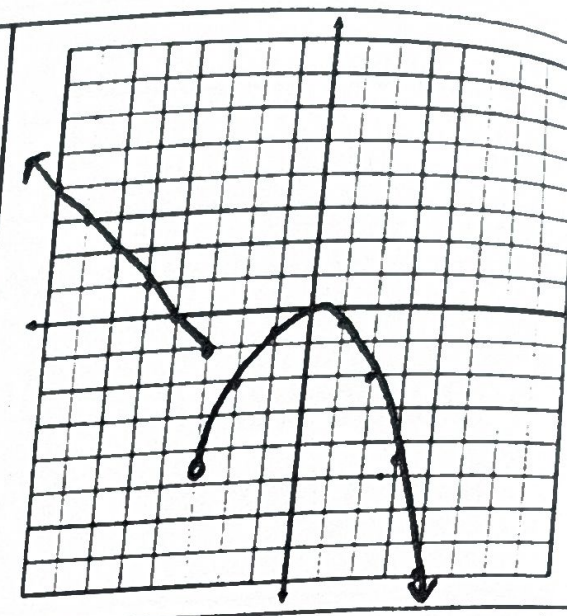
Domain: $\mathbb{R}$

Range: $\{y \mid y \ge -5\}$

Discontinuities: $x = 0$; jump



4. $h(x) = \begin{cases} -4 - x & \text{if } x \le -3 \\ -\frac{1}{2}x^2 & \text{if } x > -3 \end{cases}$

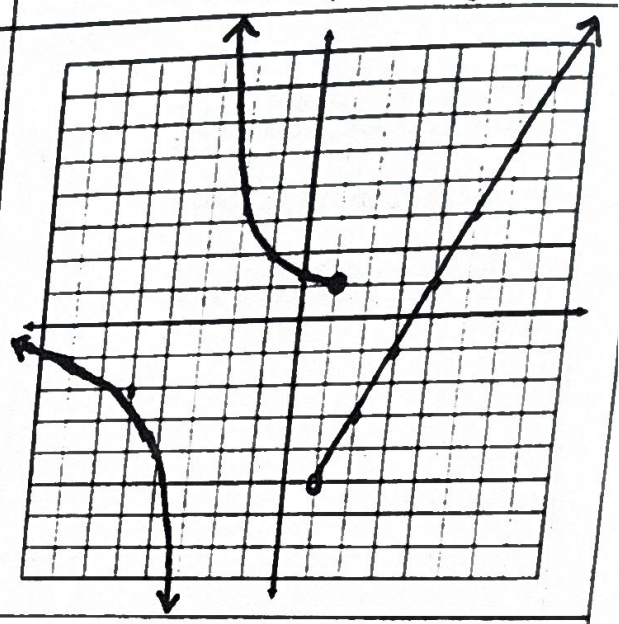


Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Discontinuities:
$x = -3$; jump

5. $f(x) = \begin{cases} \frac{4}{x+3} & \text{if } x \le 1 \\ 2x - 7 & \text{if } x > 1 \end{cases}$

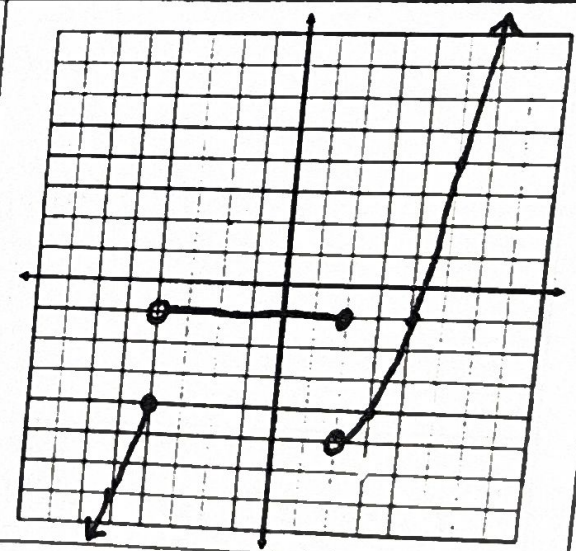


Domain: $\{x \mid x \neq -3\}$

Range: $\mathbb{R}$

Discontinuities:
$x = -3$; infinite
$x = 1$; jump

6. $g(x) = \begin{cases} 3x + 8 & \text{if } x \le -4 \\ -1 & \text{if } -4 < x \le 2 \\ (x-2)^2 - 5 & \text{if } x > 2 \end{cases}$

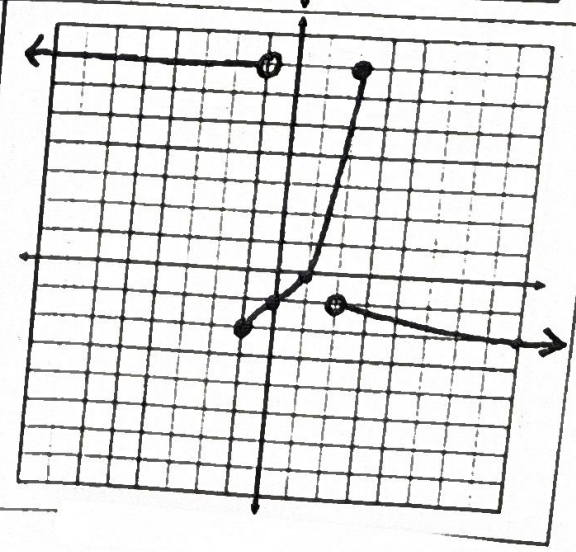


Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Discontinuities:
$x = -4$; jump
$x = 2$: jump

7. $f(x) = \begin{cases} 7 & \text{if } x < -1 \\ x^3 - 1 & \text{if } -1 \le x \le 2 \\ -\sqrt{2x} + 1 & \text{if } x > 2 \end{cases}$



Domain: $\mathbb{R}$

Range: $\{y \mid y \le 7\}$

Discontinuities:
$x = -1$; jump
$x = 2$; jump

Name: _____ **Unit 2:** Functions & Their Graphs

Date: _____ **Per:** _____ **Homework 8:** Piecewise Functions

**** This is a 2-page document! ****

Directions: Graph each function, then give the domain, range, and identify the location and type of any discountinuities.

1. $f(x)=\begin{cases} x^3-1 & \text{if } x<-2 \\ 3x+5 & \text{if } x\geq -2 \end{cases}$

a) $f(-4)$

$(-4)^3-1$

$-64-1$

$=\boxed{-65}$

b) $f(-2)$

$3(-2)+5$

$-6+5$

$=\boxed{-1}$

2. $g(x)=\begin{cases} -x^2+7 & \text{if } x\leq -7 \\ \sqrt{3x+15} & \text{if } -7<x<5 \\ 6-x & \text{if } x\geq 5 \end{cases}$

a) $g(-7)$

$-(-7)^2+7$

$-49+7$

$=\boxed{-42}$

b) $g(4)$

$\sqrt{3(4)+15}$

$\sqrt{12+15}$

$\sqrt{27}=\boxed{3\sqrt{3}}$

3. $f(x)=\begin{cases} [\![3x]\!] & \text{if } -7<x<2 \\ |x-4|+5 & \text{if } x\geq 2 \end{cases}$

a) $f(-1.2)$

$[\![3(-1.2)]\!]$

$[\![-3.6]\!]$

$=\boxed{-4}$

b) $f(2)$

$|2-4|+5$

$|-2|+5$

$2+5=\boxed{7}$

4. $h(x)=\begin{cases} -|3x-2| & \text{if } x<-5 \\ x^2-5x & \text{if } -5\leq x\leq 0 \\ \frac{3}{2}x+7 & \text{if } x>0 \end{cases}$

a) $h(-5)$

$(-5)^2-5(-5)$

$25+25$

$=\boxed{50}$

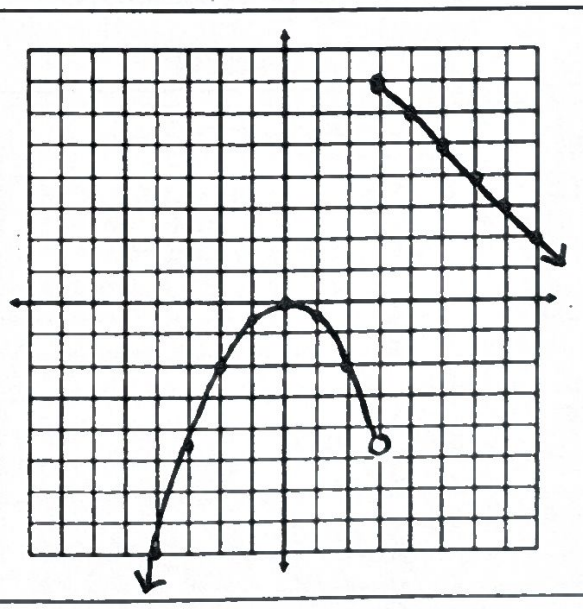
b) $h\left(\frac{8}{15}\right)$

$\frac{3}{2}\left(\frac{8}{15}\right)+7$

$\frac{4}{5}+7=\boxed{\frac{39}{5}}$ (7.8)

Directions: Graph each function, then give the domain, range, and identify the location and type of any discountinuities.

5. $f(x)=\begin{cases} -\frac{1}{2}x^2 & \text{if } x<3 \\ 10-x & \text{if } x\geq 3 \end{cases}$

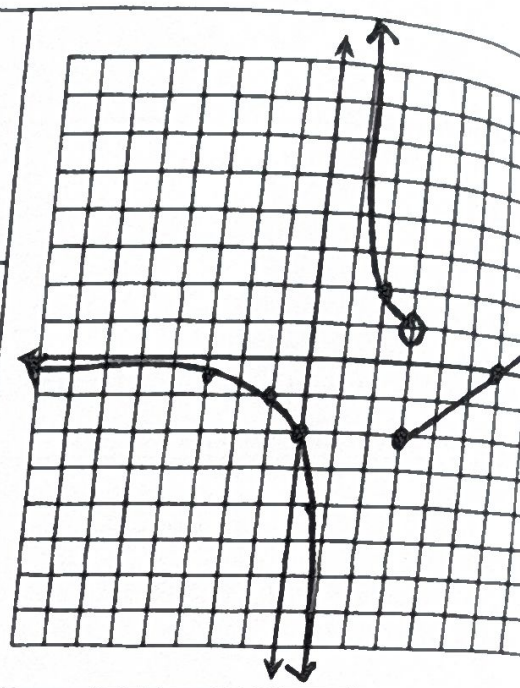


Domain: $\mathbb{R}$

Range: $\{y \mid y\leq 7\}$

Discontinuities: $X=3$; jump

6. $h(x) = \begin{cases} \frac{2}{x-1} & \text{if } x < 3 \\ \frac{2}{3}x - 4 & \text{if } x \geq 3 \end{cases}$

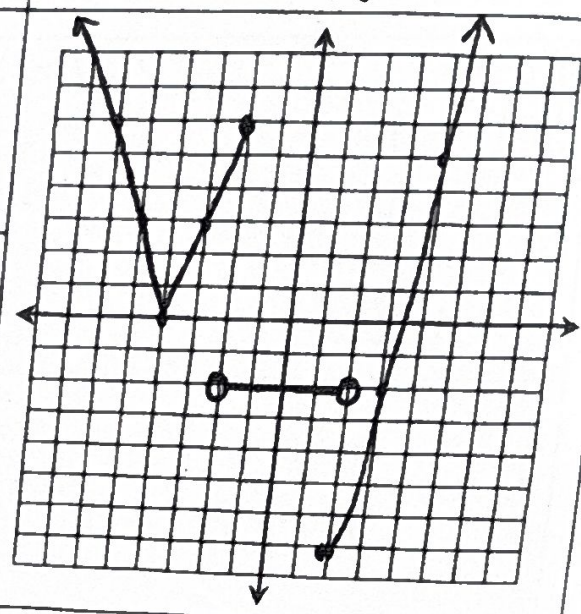


Domain:
$\{x \mid x \neq 1\}$

Range:
$\mathbb{R}$

Discontinuities:
$x = 1$; infinite
$x = 3$; jump

7. $g(x) = \begin{cases} 3|x+4| & \text{if } x \leq -2 \\ -2 & \text{if } -2 < x < 2 \\ x^2 - 11 & \text{if } x \geq 2 \end{cases}$

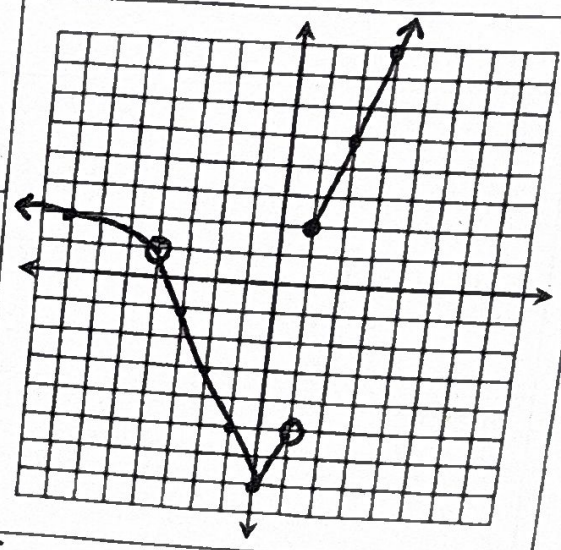


Domain:
$\mathbb{R}$

Range:
$\{y \mid y \geq -7\}$

Discontinuities:
$x = -2$; jump
$x = 2$; jump

8. $h(x) = \begin{cases} \sqrt{-(x+3)} & \text{if } x < -4 \\ |2x| - 7 & \text{if } -4 < x < 1 \\ 3x - 1 & \text{if } x \geq 1 \end{cases}$

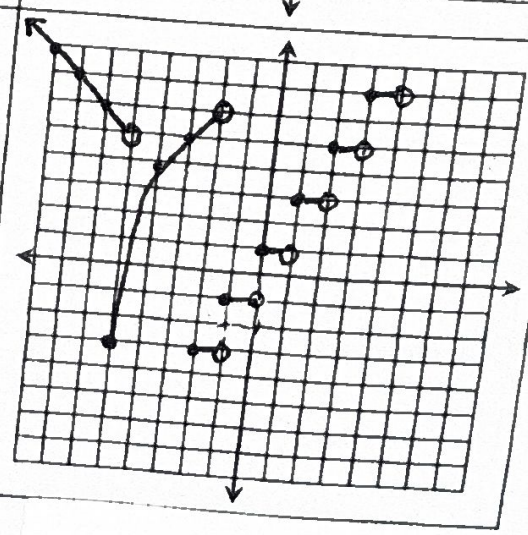


Domain:
$\{x \mid x \neq -4\}$

Range:
$\{y \mid -7 \leq y < 1 \text{ or } y > 1\}$

Discontinuities:
$x = -4$; removable
$x = 1$; jump

9. $f(x) = \begin{cases} -x & \text{if } x < -5 \\ (x+3)^3 + 5 & \text{if } -5 \leq x < -2 \\ 2[\![x]\!] + 1 & \text{if } x \geq -2 \end{cases}$



Domain:
$\mathbb{R}$

Range:
$\{y \mid y \geq -3\}$

Discontinuities:
$x = -5$; jump
$x = -2$; jump
$x = -1$; jump (and every int. after)